

NET731 – Networking

Week 7 Practical: Monitoring Setup

17–23 April | Total Marks: 10

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Submission Date:	07 May 2026
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Learning Outcomes

By the end of this practical, you should be able to:

- Enable and configure Azure Monitor on virtual machine and network resources
- Navigate the Azure Monitor interface to locate and interpret key metrics
- Identify normal versus abnormal performance baselines for CPU, network, and disk
- Produce a structured monitoring report with supporting evidence

Overview

This week you will configure Azure Monitor across the cloud environment you have been building throughout the semester. Azure Monitor is the central platform for collecting, analysing, and acting on telemetry from your Azure resources. Effective monitoring is not simply about enabling tools — it requires understanding what you are measuring and why it matters.

You are expected to make decisions, explore the interface, and think critically about what you observe. Screenshots must clearly show your own Azure environment — generic or placeholder images will not be accepted.

Before You Begin — Prerequisites

Ensure your VM from previous weeks is running and accessible.

Your Virtual Network (VNet) and Network Security Group (NSG) from Week 5 should be in place.

You will need Contributor access to your Azure resource group.

Allow 10–15 minutes for monitoring data to populate after enabling Azure Monitor.

Task 1 — Enable Azure Monitor (4 Marks)

Azure Monitor needs to be enabled on individual resources before it can collect data. This task covers both your virtual machine and the network resources in your environment.

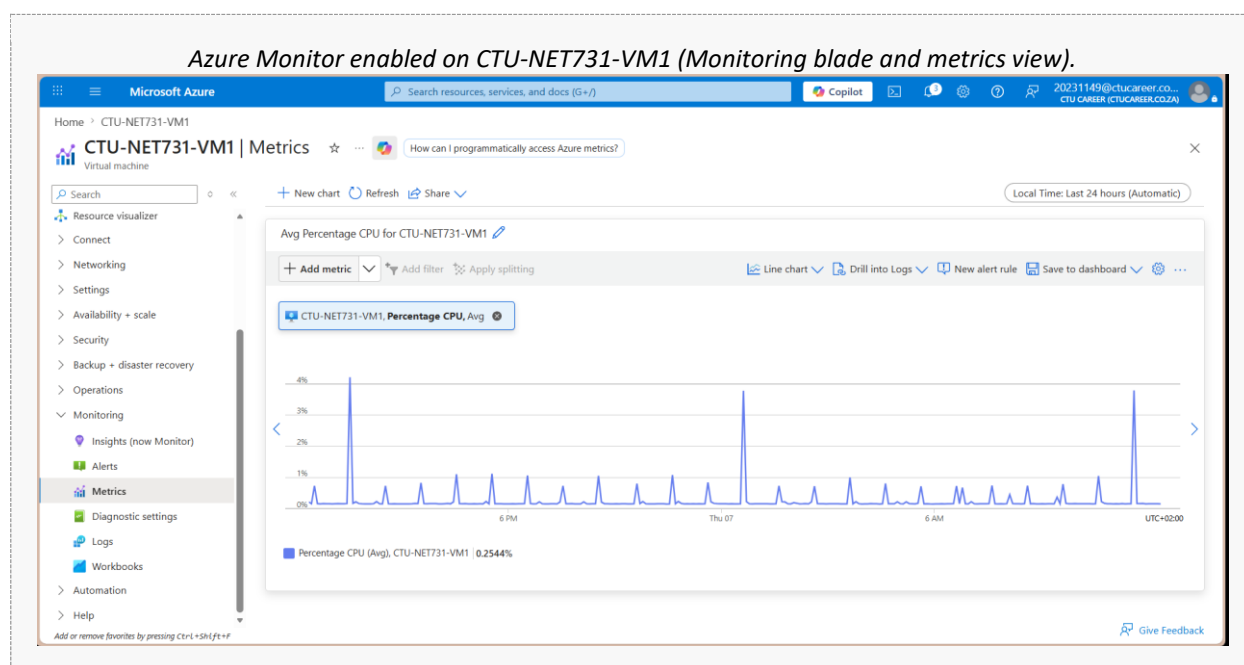
Task 1.1 — Enable Monitoring on Your Virtual Machine (2 Marks)

Navigate to your virtual machine in the Azure Portal and locate the Monitoring section in the left-hand menu. You will find several sub-options here — explore them before enabling anything.

Think Before You Click

What is the difference between 'Insights' and 'Diagnostics settings' for a VM?
What data will be collected once you enable monitoring? Where is it stored?
Does enabling monitoring have any cost implications you should be aware of?

Once you have explored the options, enable VM Insights or Diagnostics for your virtual machine. Configure it to collect at minimum: CPU metrics, memory (if supported), and disk I/O.



Describe in 2–3 sentences what you enabled and where the collected data will be stored:

Your response:

I enabled Azure Monitor on CTU-NET731-VM1 and used the Monitoring section to view CPU, disk and other VM performance data. I then used Azure Metrics Explorer to confirm that the VM was collecting live telemetry. The data can be viewed in Azure Monitor, and the network diagnostic data was also sent to the CTU-20231149-LAW Log Analytics Workspace for central storage and analysis.

Task 1.2 — Enable Monitoring on Network Resources (2 Marks)

Repeat the process for your network resources. Both your NSG and your VNet support diagnostic settings.

Consider the following before proceeding:

- What specific log categories are available for an NSG? Which ones would be most useful in a security context?
- What is a Log Analytics Workspace, and why might you need one to proceed?
- If you do not already have a Log Analytics Workspace, you will need to create one — consider which region is appropriate.

For the network resources, I enabled diagnostic settings on both the NSG and the VNet. The NSG collected Network Security Group Event and Network Security Group Rule Counter logs, while the VNet collected VM protection alerts and AllMetrics. I also created a Log Analytics Workspace called CTU-20231149-LAW in South Africa North and used it as the destination for the diagnostics.

Diagnostic settings enabled on CTU-NET731-NSG with selected categories and Log Analytics destination.

Home > CTU-NET731-NSG | Diagnostic settings

Diagnostic setting ...

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#) [JSON View](#)

Diagnostic setting name CTU-NET731-NSG-Diagnostics

Logs

Category groups

☐ allLogs

Categories

☒ Network Security Group Event

☒ Network Security Group Rule Counter

Destination details

☒ Send to Log Analytics workspace

Subscription

Azure for Students

Log Analytics workspace

CTU-20231149-LAW (southafricanorth)

☐ Archive to a storage account

☐ Stream to an event hub

☐ Send to partner solution

Diagnostic settings enabled on CTU-NET731-VNet with selected categories, metrics and Log Analytics destination.

Home > Network foundation | Virtual networks > CTU-NET731-VNet | Diagnostic settings

Diagnostic setting ...

Save Discard Delete Feedback

A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. [Learn more about the different log categories and contents of those logs](#) [JSON View](#)

Diagnostic setting name CTU-NET731-VNet-Diagnostics

Logs

Category groups

☐ allLogs

Categories

☒ VM protection alerts

Metrics

☒ AllMetrics

Destination details

☒ Send to Log Analytics workspace

Subscription

Azure for Students

Log Analytics workspace

CTU-20231149-LAW (southafricanorth)

☐ Archive to a storage account

☐ Stream to an event hub

☐ Send to partner solution

Task 2 — Review Key Metrics (3 Marks)

Now that monitoring is enabled, you will use Azure Monitor's Metrics Explorer to observe and interpret the data being collected. Generating some load on your VM before taking screenshots will produce more meaningful graphs.

Generating Test Load (Optional but Recommended)

On Linux VMs: the 'stress' tool can simulate CPU and I/O load.

On Windows VMs: Task Manager's 'New Task' can run a CPU-intensive process.

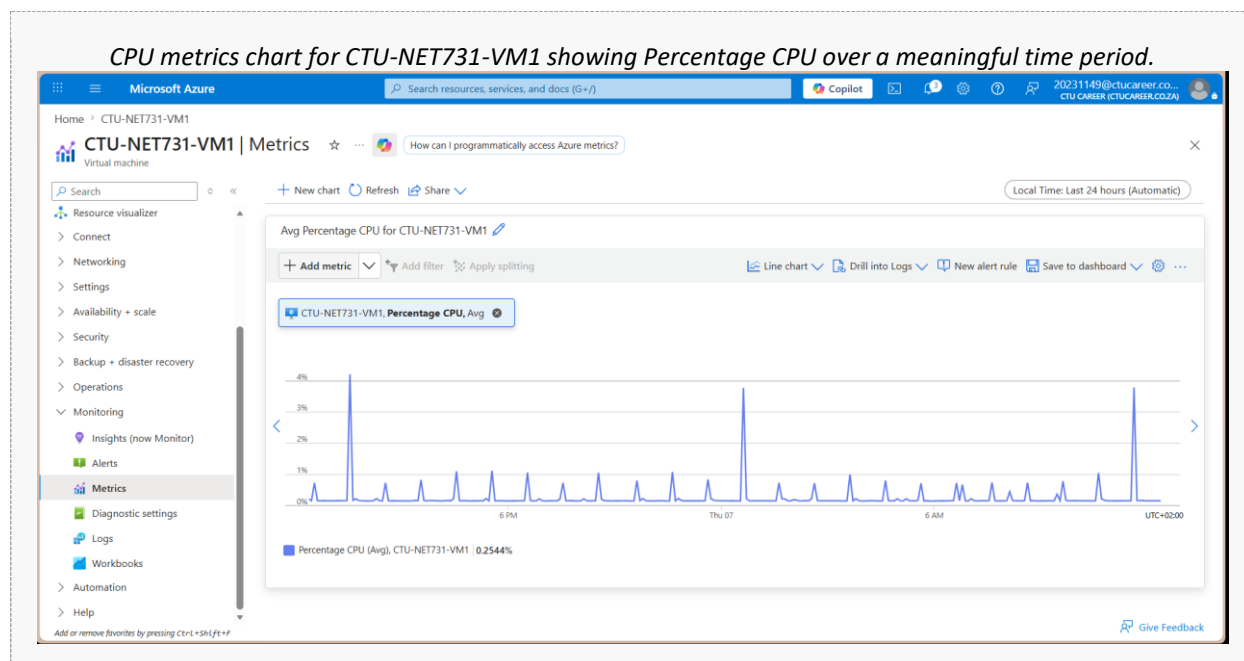
Even browsing the web or installing packages from within the VM will generate measurable network activity.

Allow a few minutes after generating load before capturing screenshots.

Task 2.1 — CPU Metrics (1 Mark)

Open Azure Monitor > Metrics and select your VM as the scope. Explore the available metrics before choosing one to display.

- Select an appropriate time range that shows meaningful data (consider: what time range makes sense for a lab exercise?)
- Identify which metric reflects actual CPU usage — consider why there might be more than one CPU-related metric
- Add annotations or use the pin feature to highlight any interesting patterns you observe



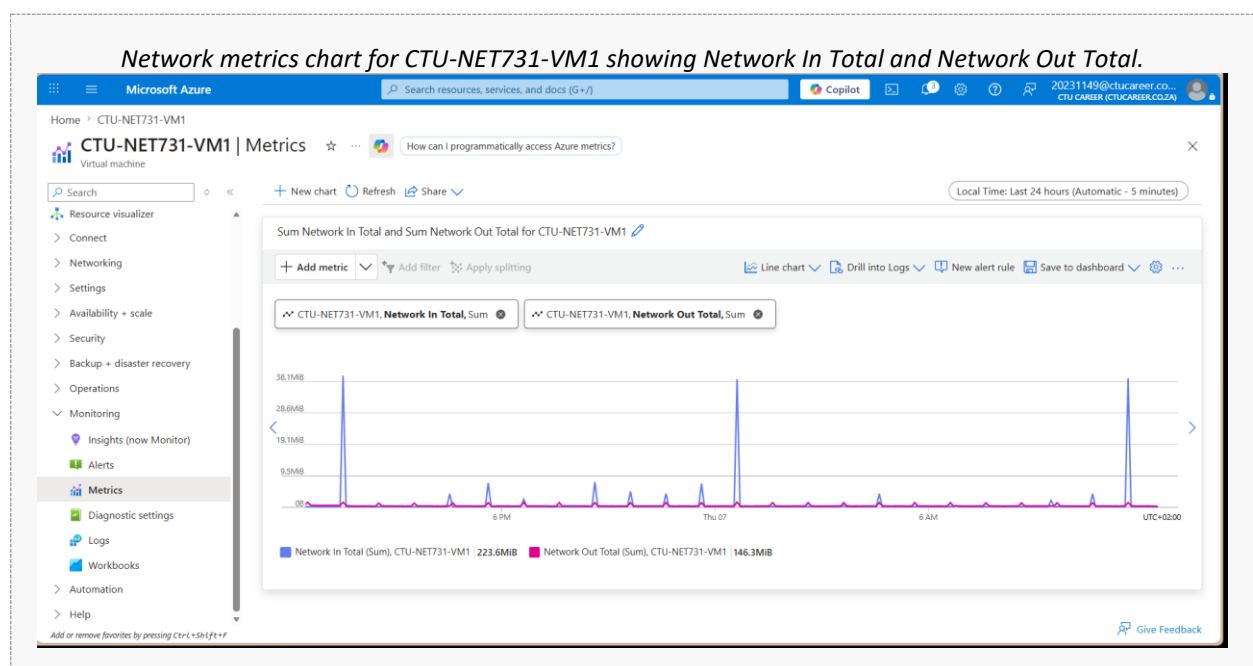
In 1–2 sentences, describe what the CPU metric is showing and whether the values appear expected for your workload:

The CPU chart shows that CTU-NET731-VM1 had low CPU usage most of the time with a few short spikes. That is expected for my workload because the VM was mainly used for lab setup, portal activity and monitoring tasks rather than heavy processing.

Task 2.2 — Network Metrics (1 Mark)

Still in Metrics Explorer, change the metric to observe network throughput. You should investigate both inbound and outbound traffic.

- What is the unit of measurement for the network metric you selected? Is it what you expected?
- Can you display both Network In and Network Out on the same chart? How would you do this?



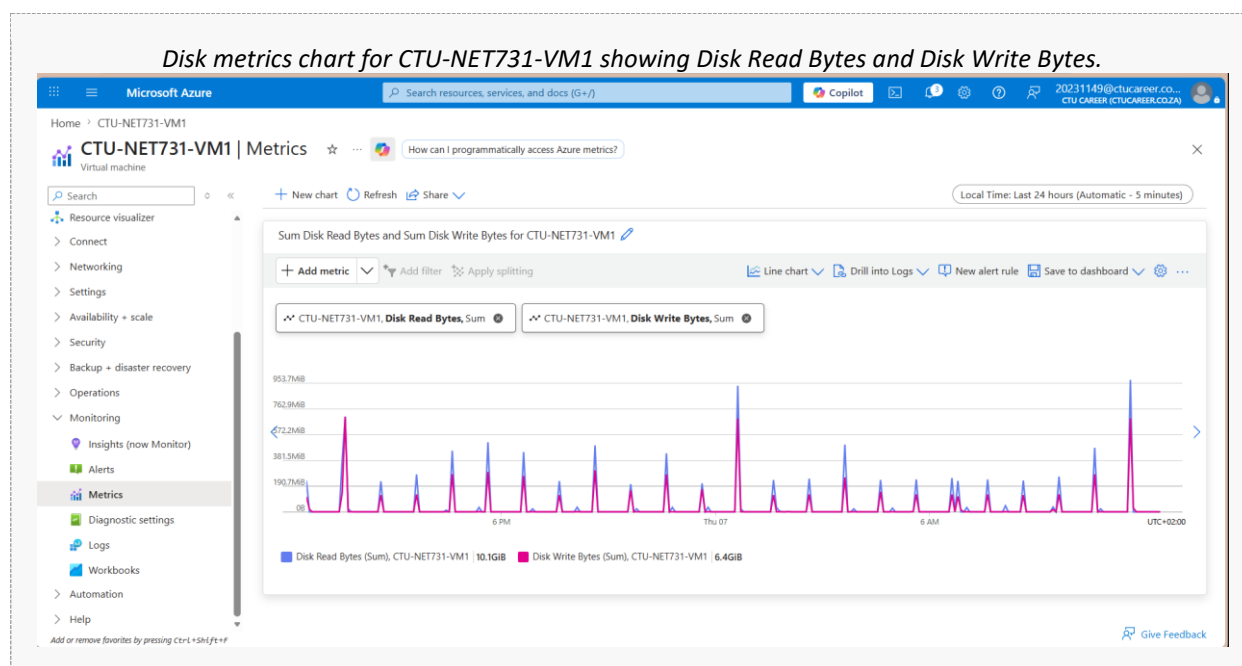
Note any observations about the network traffic pattern you can see:

The network metrics show both inbound and outbound traffic on the VM over the selected period. The traffic stays low most of the time with a few visible spikes, which is normal for a lab environment and was likely caused by Azure portal activity, background communication and management tasks.

Task 2.3 — Disk Metrics (1 Mark)

Finally, explore the disk-related metrics for your VM. Azure exposes separate metrics for disk reads and disk writes.

- Locate both Disk Read Bytes and Disk Write Bytes (or equivalent metrics in your VM configuration)
- Consider: during what activities would you expect disk write activity to be high?



Describe in 1–2 sentences what the disk metrics reveal about your VM's activity:

The disk metrics show repeated read and write bursts instead of constant heavy usage. This suggests normal operating system activity, monitoring activity and short background processes rather than a sustained workload.

Task 3 — Document Your Monitoring Setup (3 Marks)

Monitoring documentation is a critical deliverable in professional environments. A future administrator — or your future self — should be able to understand your setup from the document alone.

Task 3.1 — Monitoring Summary (2 Marks)

Write a structured summary of your monitoring configuration. Your summary should cover all three areas below. Aim for clarity and precision — this is professional technical documentation.

Resources Monitored

List each Azure resource you have enabled monitoring on, specifying the monitoring method used (e.g., VM Insights, Diagnostic Settings) and what data each resource is collecting:

Monitoring was enabled on CTU-NET731-VM1 using Azure Monitor and the Metrics blade, which allowed me to review CPU, network and disk activity. Monitoring was also enabled on CTU-NET731-NSG through Diagnostic Settings, where Network Security Group Event and Network Security Group Rule Counter were selected. CTU-NET731-VNet was also configured with Diagnostic Settings, using VM protection alerts and AllMetrics as the selected categories.

Data Retention and Storage

Where is your monitoring data stored? What retention period applies? How would you access historical data older than the default retention window?

The NSG and VNet diagnostic data was configured to go to the CTU-20231149-LAW Log Analytics Workspace. I did not manually change the retention setting during the practical, so the workspace keeps the default retention configured for that workspace. Older monitoring data can be accessed later through Azure Monitor and the Log Analytics Workspace by running queries and reviewing saved logs and metrics.

Metric Observations

Summarise what you observed across the three metrics reviewed in Task 2. Were any values unexpected? What might cause anomalies in these metrics in a production environment?

The CPU graph showed low normal usage with a few small spikes, which was expected. The network graph showed both inbound and outbound traffic with occasional bursts, and the disk graph showed regular read and write spikes from normal VM activity. In a production environment, unusual spikes in these metrics could be caused by high user load, software updates, backups, malware, scheduled jobs or performance bottlenecks.

Task 3.2 — Extension: Configure an Alert Rule (1 Bonus Mark)

Extension Task

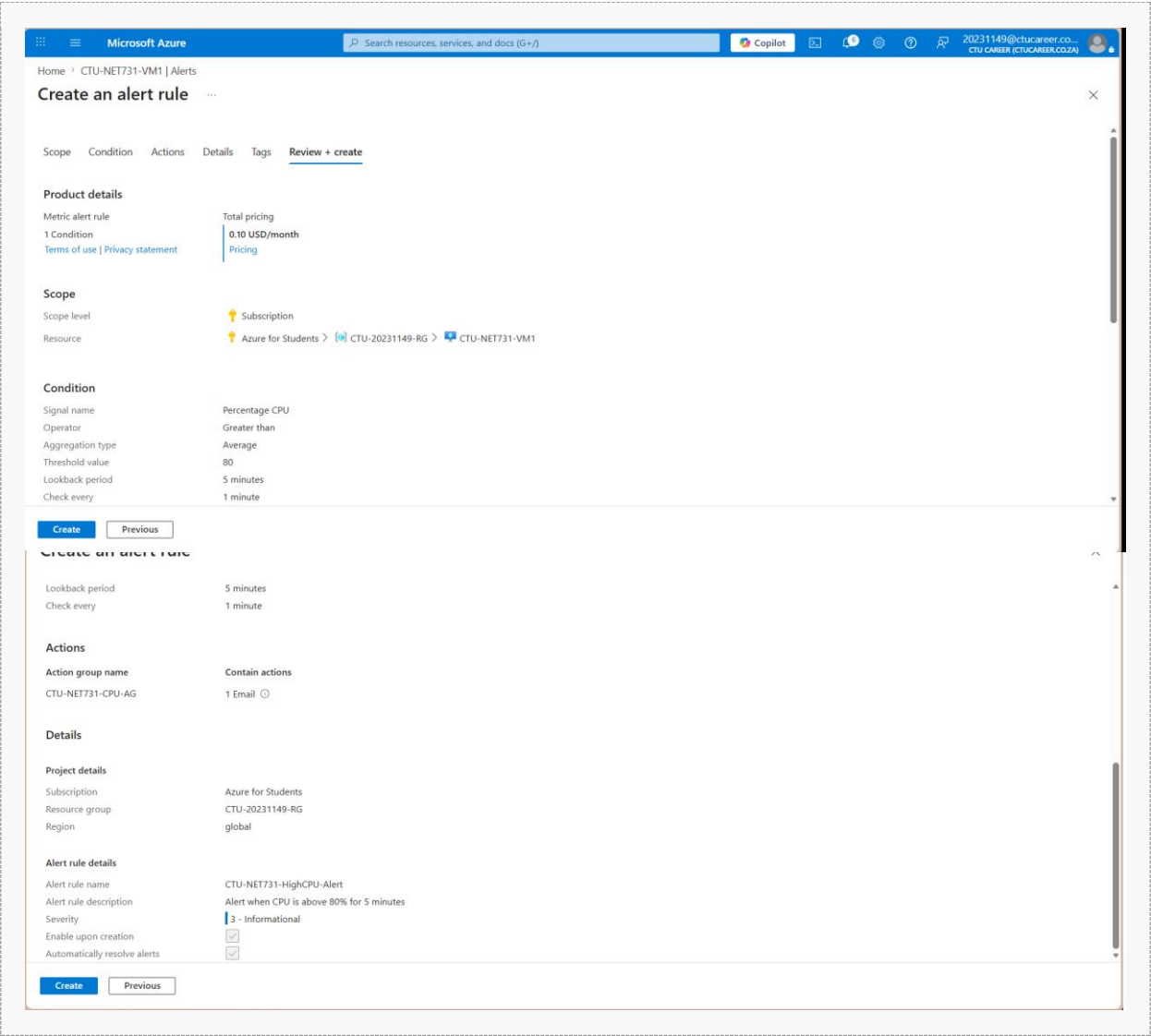
This task is optional but provides an opportunity to demonstrate deeper understanding.

Configure at least one Azure Monitor Alert Rule on your VM or network resource.

The alert should have a meaningful threshold — for example, triggering when CPU exceeds 80% for 5 minutes.

Document your alert logic: what condition triggers it, what action does it take, and why you chose that threshold.

Azure Monitor alert rule configured for high CPU usage with condition, threshold and action group.



Explain the alert rule you configured and the reasoning behind your chosen threshold:

I configured an Azure Monitor alert rule on CTU-NET731-VM1 to trigger when Percentage CPU is greater than 80% for 5 minutes. The alert uses the CTU-NET731-CPU-AG action group to send an email notification. I chose this threshold because sustained CPU usage above 80% can point to abnormal workload or a performance issue, while still avoiding alerts for very small spikes.

Reflection Questions

Answer these questions thoughtfully. There are no single correct answers — markers are looking for evidence of understanding and critical thinking.

Reflection Question	Your Response
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Why is monitoring important in a cloud environment? How does it differ from monitoring on-premises infrastructure?	Monitoring is important in the cloud because it helps you track performance, availability and security across resources that can change quickly. In Azure, the monitoring tools are built into the platform and are more centralised, while on-premises monitoring often depends on separate local tools and more manual setup.
What is the difference between a metric and a log in Azure Monitor? When would you use each?	A metric is a numerical measurement collected over time, such as CPU percentage, network traffic or disk bytes. A log is a record of an event or activity, such as a security event or a diagnostic entry. I would use metrics for graphs, trends and alerts, and logs for deeper investigation and troubleshooting.
If your VM showed consistently high CPU usage (above 90%) every day at 2am, what steps would you take to investigate?	I would first confirm the pattern in Azure Monitor and check whether the CPU spike happens at the same time every day. After that, I would investigate scheduled tasks, backups, software updates, antivirus scans, scripts or other processes running at 2am. I would also compare CPU with disk and network metrics to see whether another activity is causing the spike.
How could monitoring data be used to make cost-optimisation decisions in Azure?	Monitoring data can help show whether a VM is underused or overused. If CPU, network and disk activity stay low for long periods, the resource may be oversized and could be changed to a cheaper option. Monitoring can also help identify idle resources, unnecessary services and good times to use auto-shutdown or scheduling.

Marking Criteria

Use this table to check your submission before submitting.

Task	Marks	Evidence Required
1.1 Enable Azure Monitor on VM	2	<i>Screenshot of Monitor enabled on VM resource</i>
1.2 Enable Monitoring on Network Resources	2	<i>Screenshot showing NSG/VNet diagnostics</i>
2.1 CPU Metrics Review	1	<i>Screenshot of CPU usage chart with annotations</i>
2.2 Network Metrics Review	1	<i>Screenshot of network in/out metrics</i>
2.3 Disk Metrics Review	1	<i>Screenshot of disk read/write metrics</i>

3.1 Monitoring Documentation	2	<i>Written summary with all screenshots included</i>
3.2 Alert Configuration (Extension)	1	<i>Screenshot of configured alert rule</i>
TOTAL	10	

Submission Checklist

All screenshot placeholders have been replaced with real screenshots from your Azure environment.

Each screenshot clearly shows your resource names and Azure subscription context.

Written responses are your own words — not copied from documentation.

Reflection questions have been answered with evidence of understanding.

The document has been saved as a .docx or .pdf and submitted via the course portal.

Useful References

The following official documentation may help — but always explore the Azure Portal first before consulting documentation:

- Azure Monitor overview — docs.microsoft.com/azure/azure-monitor/overview
- Enable VM Insights — docs.microsoft.com/azure/azure-monitor/vm/vminsights-enable-overview
- Metrics Explorer — docs.microsoft.com/azure/azure-monitor/essentials/metrics-getting-started
- Diagnostic settings — docs.microsoft.com/azure/azure-monitor/essentials/diagnostic-settings
- Azure Monitor Alerts — docs.microsoft.com/azure/azure-monitor/alerts/alerts-overview

Tip: The Azure Portal's built-in 'Learn more' links on each blade are often the fastest route to relevant documentation.

Log Analytics Workspace Created

The CTU-20231149-LAW Log Analytics Workspace was created in South Africa North and used as the destination for the NSG and VNet diagnostic settings.

Successful deployment of the CTU-20231149-LAW Log Analytics Workspace.

The screenshot displays the Microsoft Azure portal interface. At the top, the header shows 'Microsoft Azure' and a search bar. Below the header, the 'Home' section is visible, followed by the 'Microsoft.LogAnalyticsOMS | Overview' page. The page title is 'Deployment'. On the left sidebar, there are tabs for 'Overview', 'Inputs', 'Outputs', and 'Template'. The main content area shows a green checkmark icon and the text 'Your deployment is complete'. Below this, deployment details are listed: 'Deployment name : Microsoft.LogAnalyticsOMS', 'Subscription : Azure for Students', and 'Resource group : CTU-20231149-RG'. The 'Start time' is '07/05/2026, 12:13:36' and the 'Correlation ID' is 'c99bbdb0-1249-4c01-ab9e-f5274bac0902'. A table titled 'Deployment details' shows a single resource: 'CTU-20231149-LAW' of type 'Log Analytics workspace' with a status of 'OK'. Below the table, there is a 'Next steps' section with a 'Go to resource' button. On the right sidebar, there are several informational cards: 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'. The URL at the bottom of the browser window is 'https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~fin...'

References

- Microsoft. (2026). Azure Monitor overview. Microsoft Learn.
- Microsoft. (2026). Diagnostic settings in Azure Monitor. Microsoft Learn.
- Microsoft. (2026). Metrics Explorer in Azure Monitor. Microsoft Learn.
- Microsoft. (2026). Azure Monitor alerts overview. Microsoft Learn.
- Microsoft. (2026). Log Analytics workspace overview. Microsoft Learn.